

**"One identity is never
enough to explain
AI harm."**

The human experience is intersectional. Our audits must be too

RESEARCH DEFENSE 2026

Humans Are Not Just One Thing

Systematic Study of Intersectional AI Harms in Workplace Settings

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| Beyond the single axis



Standard Fairness

Tests one protected attribute at a time.



Intersectionality

Crenshaw: identities interact and overlap.



Our Objective

Apply a strict causal filter to workplace AI.

| The Crisis of Single-Axis Audits

FLAWED ASSUMPTION



The Additive Fallacy

Compound risk is not a simple sum.

DATA LIMITATION



The Causality Gap

Databases rarely link harm to specific causal identities.

| What Are We Asking?

RQ1

Which **identity categories** appear most often in workplace AI incidents?

RQ2

Which **identity intersections** amplify harm the most?

| What Are We Asking?

RQ3

Do media outlets **explicitly report** intersectional harms?

RQ4

Are **oppressed groups** covered more frequently in news than privileged groups?

| Data Collection & Filtering

STEP 01



Global Sources

Started with **58,000** raw AI incidents.

STEP 02



Context Filter

Strictly isolated **workplace** contexts.

STEP 03



Final Dataset

Curated **285** unique human incidents.

| Extraction & Validation

CAUSAL FRAMEWORK

CQ1: Direct Impact

Did identity *cause* the incident?

CQ2: Necessity

Would it happen *without* the identity?

IMPLEMENTATION & RIGOR

Gemini Flash API

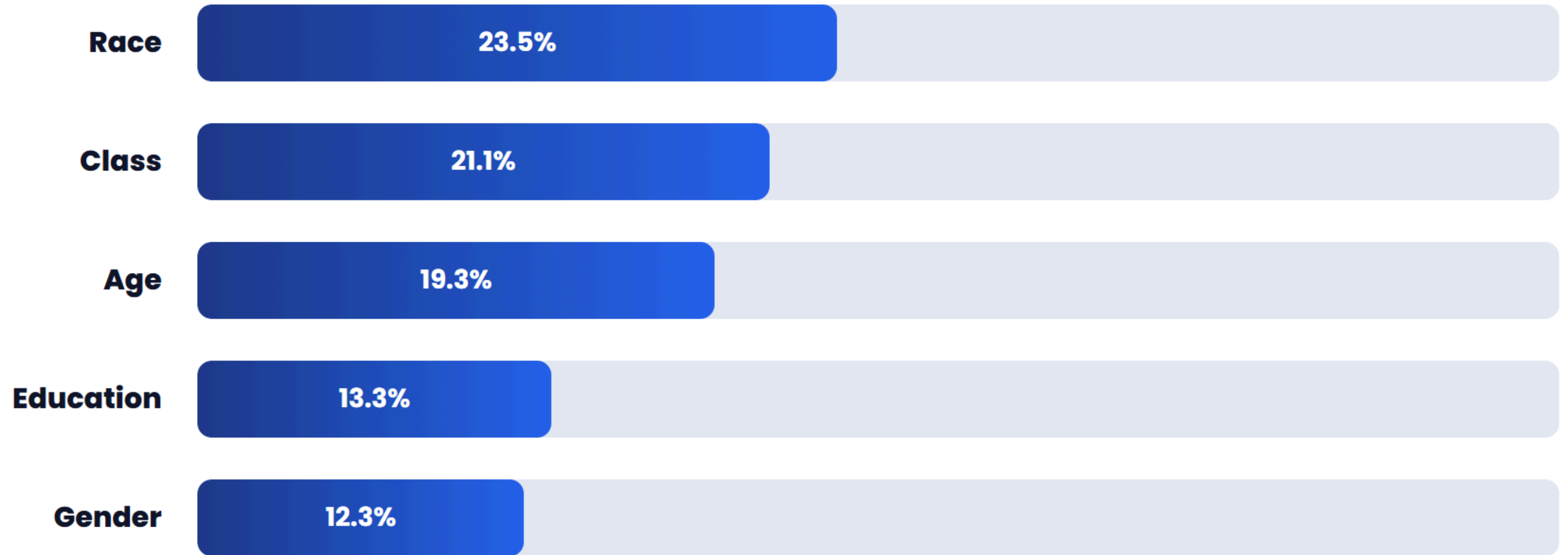
Native JSON + strict counterfactual logic.

Kept only if: CQ1 = Yes & CQ2 = No

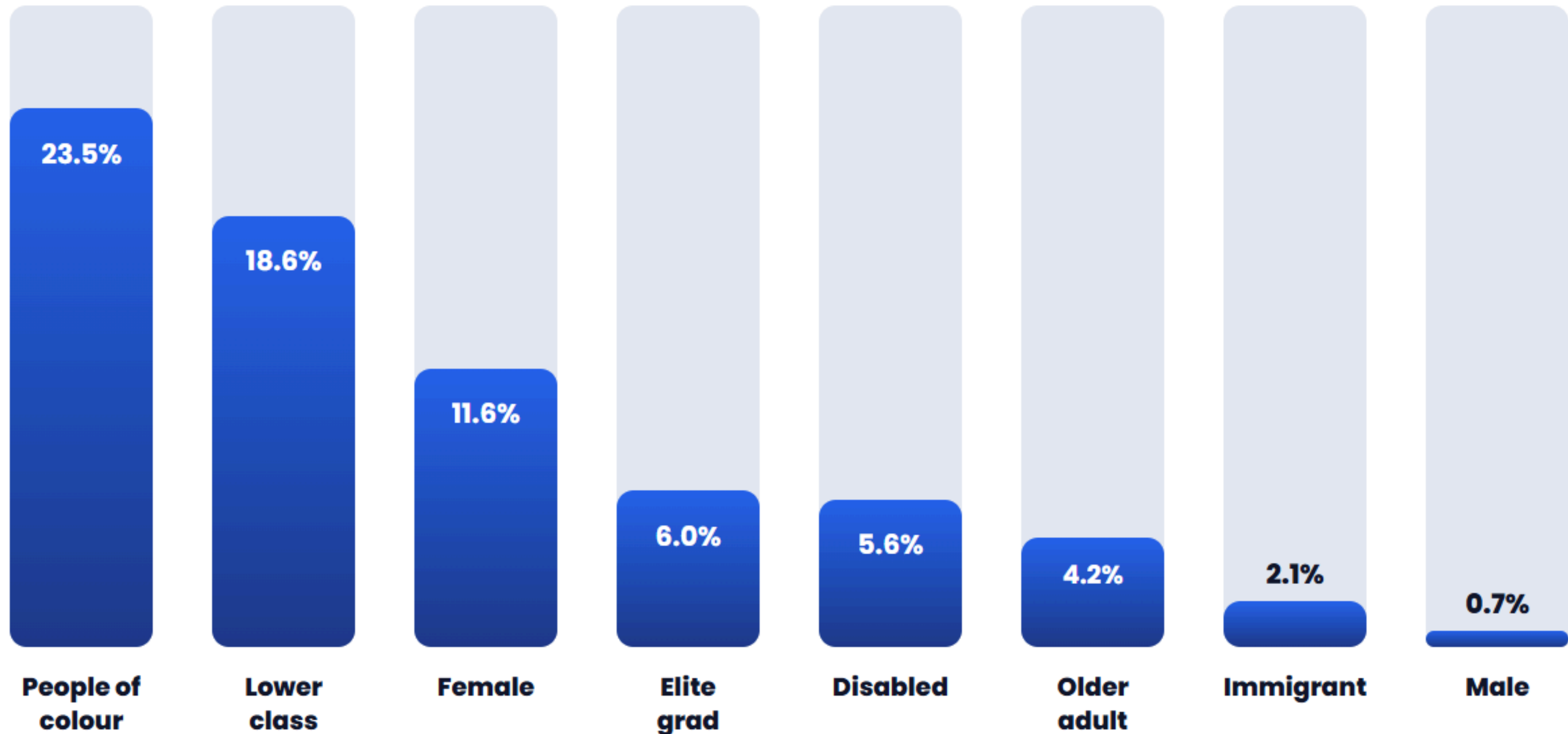
Human Validation

10% randomly checked ($\kappa = 0.81$).

| RQ1 — Leading Identity Categories



RQ1 — Identity Values



| RQ2 — Intersectional Spots

12

Age × Education

Co-occurring incidents

11

Age × Race

Co-occurring incidents

8

Gender × Race

Co-occurring incidents

8

Gender × Age

Co-occurring incidents

| RQ2 — The Most Harmed Profiles

**Lower Class +
People of Colour**

6

**Female +
People of Colour**

4

**Disabled +
Older Adult**

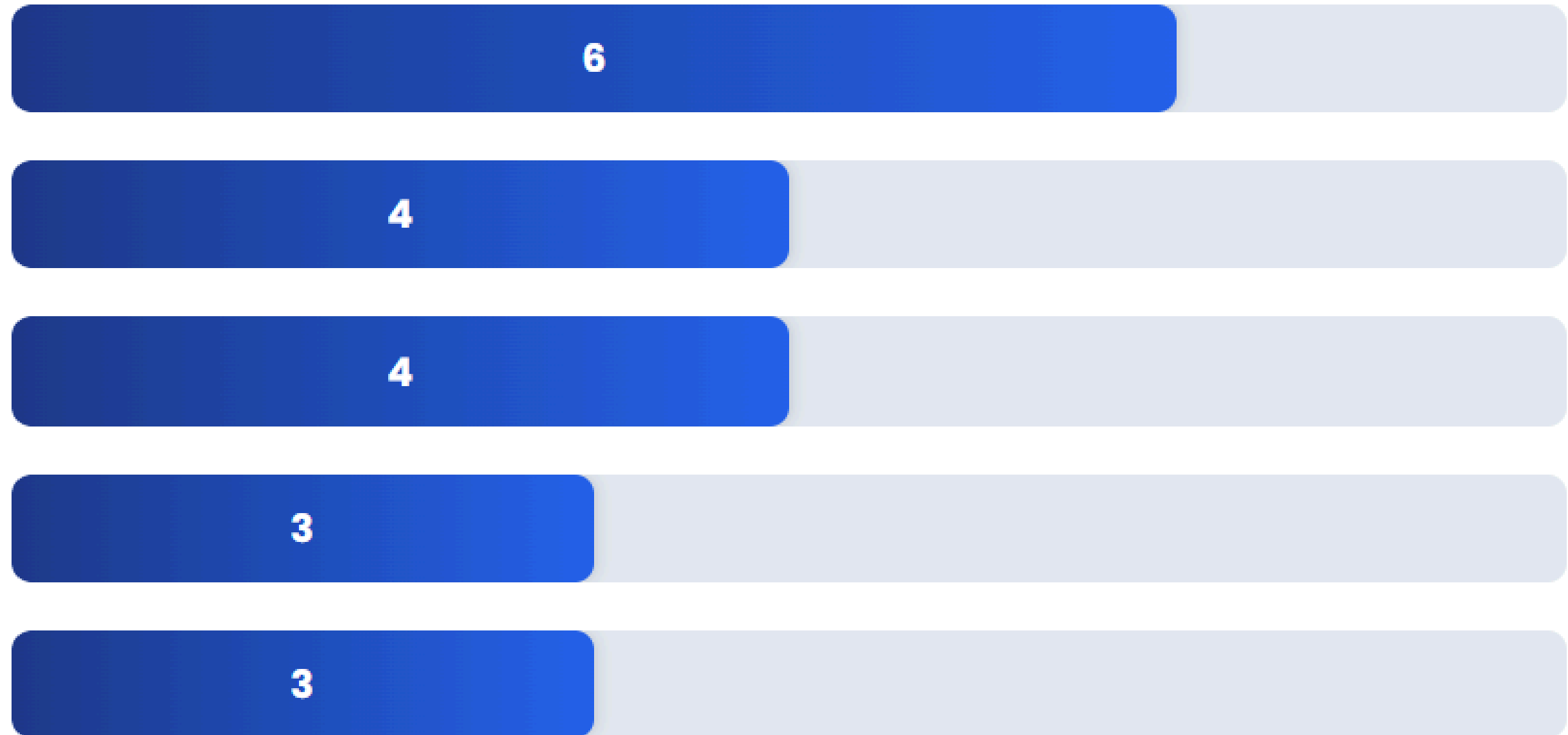
4

**Young Adult +
People of Colour**

3

**Lower Class +
Single Parent**

3



| RQ2 — Intersectional Spots

5.94x

Risk Multiplier

Disabled + Older Adult

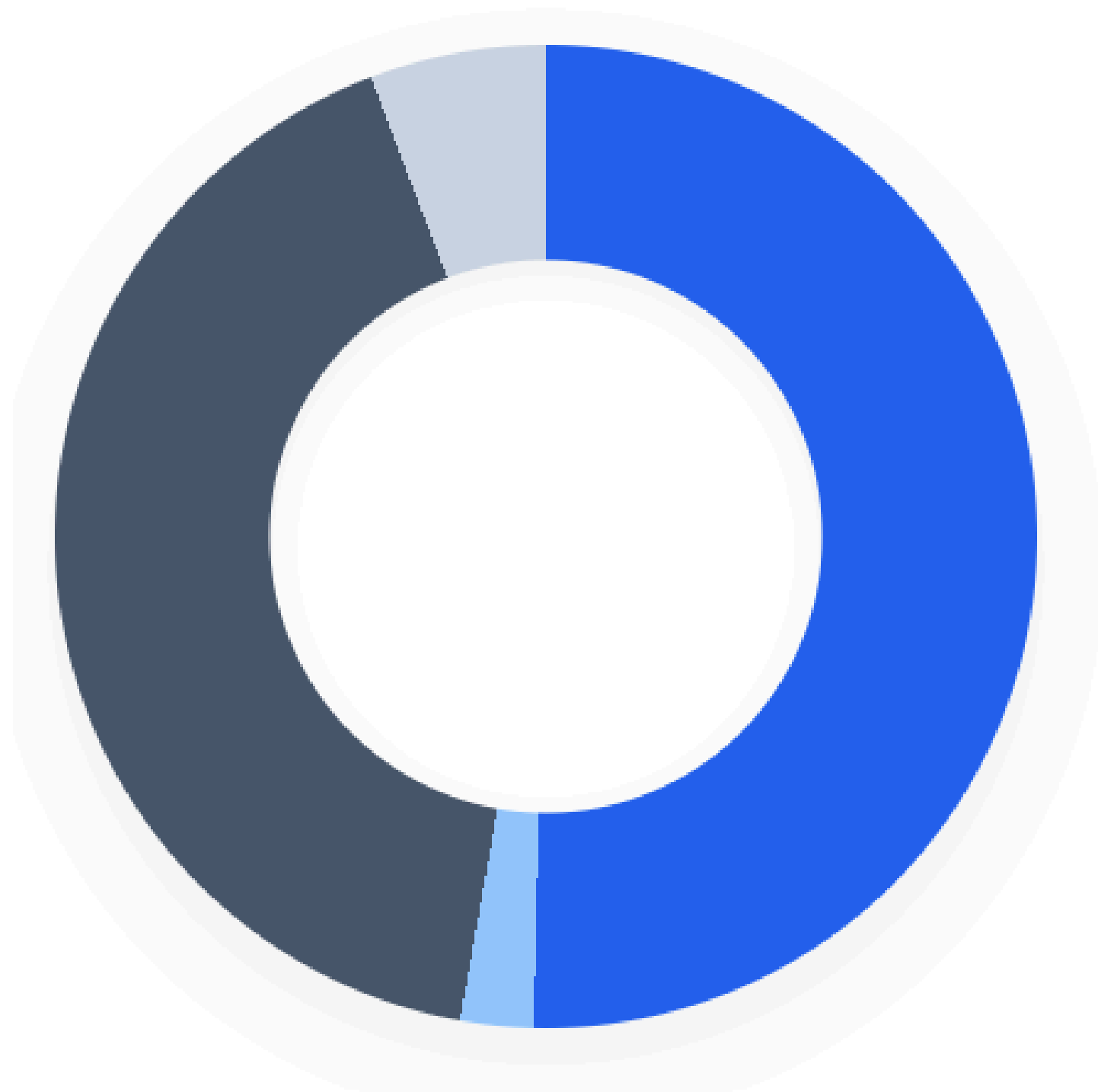
Compounded Vulnerability

Single Parent + Female **5.52x**

Adolescent + Dark Skin **3.39x**

Lower Class + Single Parent **3.35x**

| RQ3 — The Media Erasure Index



EXPLICITLY NAMED (52.8%)

	Explicit Oppressed	50.4%	(n=207)
	Explicit Privileged	2.4%	(n=10)

INFERRED / HIDDEN (47.2%)

	Inferred Oppressed	41.4%	(n=170)
	Inferred Privileged	5.8%	(n=24)

| RQ3 — The Invisibility of Class

Inferred as Causal

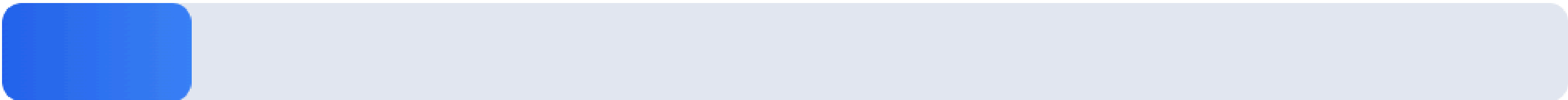
104

Incidents



Explicitly Named by Media

10



| RQ4 — Oppressed vs Privileged Coverage

86.4%

Oppressed Groups
of all covered identity markers

Vice.com

Marginalized Subject Coverage

100%

ProPublica

Marginalized Subject Coverage

100%

| RQ3 & RQ4 — Media Bias

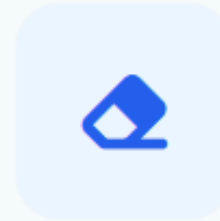
RQ4 • REPRESENTATION



The Reporting Gap

86.4% of markers belong to oppressed groups.

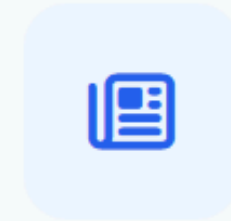
RQ3 • ERASURE



Simplification Trap

High visibility $\neq$ intersectional depth
(E=0.472).

MEDIA LANDSCAPE



Outlet Distribution

Guardian **86%**, NYT **89%**
High coverage, low depth.

| Discussion & Limitations



The Class Blind Spot

No legal protection for economic class.



Divergent Amplification

Up to **5.94x** risk in specific pairs.



Data Sparsity (N=285)

Small dataset, many identity pairs with $n < 3$.



Methodological Constraints

Assuming statistical independence between categories.

| Conclusion: Fixing the Blind Spot

POINT 01

A Structural Reality

Workplace AI harms are rarely confined to a single demographic axis; they emerge through overlapping social structures.

POINT 02

The Paradigm Shift

We must transition from basic, single-axis fairness metrics to true intersectional technical auditing.

POINT 03

A New Reporting Standard

We need a new reporting standard that names identity categories with the same rigour as race and gender.

**"Algorithmic harm does
not add up. It multiplies"**